Synthetic Cultural Agents as Measurement Instruments: A Preregistered Evaluation from Cross-National Anchor to Survey Respondents

Abstract

When a method conditioning a language model on a culture preserves an external benchmark, what has been shown about survey respondents? We study one method. Adapters are compiled from six declared country-level signs of a preference benchmark, through synthetic preference pairs, and read survey items without naming the country. Training installs the synthetic labels on unseen pairs, and on trust the adapters keep the benchmark's order of countries across a new scenario bank. A prompt stating the same signs does too, so the signs carry it. The dimensions are entangled. A preregistered screen of held-out items, against a retrospective criterion, yields a weak signal that registered contrasts cannot separate from questionnaire readers. The adapters' order of countries does not detectably agree with respondents, and their answer distributions lie farther from the survey than the base model's; a country-named prompt lies closer. The instrument can complement human pretesting; pre-piloting remains untested.

Keywords: synthetic respondents; large language models; measurement validity; survey item screening; preregistration; cross-national measurement.

1 Introduction

A method that conditions a language model on a culture is usually judged by whether it reproduces something external, such as a published ranking of countries or the answers of a survey. Suppose it does. What has then been shown about the people who answer survey questions, and about survey items that nobody has fielded? A retained benchmark is evidence about the link between benchmark and method. It is evidence about the link between method and respondents only if a further test says so, and the two links can fail independently. This paper puts that question to one method and reports, link by link, what a retained benchmark warrants.

The question has practical weight. Cross-national survey programs fix their items long before they see how those items behave: a question must be translated, pretested and fielded in many countries before a researcher learns whether its country-level differences follow the construct. Language models offer a cheap first reading of such items (Argyle et al., 2023; Horton et al., 2023; Buskirk et al., 2026), and a method that keeps a benchmark's order of countries invites the inference that it can be trusted to read new items. The literature on synthetic respondents is mostly cautionary about that step. Synthetic responses are sensitive to prompt wording, compress the variance of human answers, and depend on the order and labelling of answer options (Bisbee et al., 2024; Boelaert et al., 2025; Wang et al., 2025; Dominguez-Olmedo et al., 2024), and alignment procedures change how faithfully models simulate attitudes (Lyman et al., 2025). Methodological work in this journal concludes that validation against human data remains essential (Kozlowski and Evans, 2025; Broska et al., 2025). Most empirical work asks whether a model reproduces answers that people already gave, whether it is prompted (Santurkar et al., 2023; Durmus et al., 2024) or fine-tuned on survey response distributions (Cao et al., 2025; Suh et al., 2025). In the fine-tuned

case the human survey sits inside the training signal. Language-model pretesting has been studied as flaw detection in question wording (Sturgis et al., 2026). Whether a method can say which new items will separate countries in an intended direction is a different test, and it needs a decision rule fixed before the human criterion is read.

We evaluate a method built for that test along a chain of three links. The anchor is a declared country-level benchmark, the Global Preferences Survey (GPS; Falk et al., 2018), whose scores on six preference dimensions supply the signs that define what the instrument is meant to carry. The instrument is a set of parameter adapters trained on synthetic response pairs that those signs label. The respondents are the people who answered Wave 7 of the World Values Survey (WVS). Treating each link as a separate measurement question is the lens proposed for evaluating generative systems in general (Wallach et al., 2025); prompted models have been compared with the GPS across countries and found to be weak proxies for it (Chen et al., 2026). We apply the lens to an instrument trained on the signs of that benchmark, under a preregistered protocol.

Three features of the construction make it informative for this question, and each is a choice that a researcher can inspect. First, its inputs are declared aggregate signs: the researcher states which six country-level moments enter and applies a sign rule that turns them into labels, so nothing about the conditioning is hidden in a country name or in human answers. Second, its training data are synthetic preference pairs. Each country's sign profile labels one shared bank of generated response pairs, and Direct Preference Optimization (Rafailov et al., 2023), read as a regularized Bradley–Terry pairwise choice model, fits one adapter per profile. No human survey answer enters training, so a human survey is a criterion the method has never seen, which is not true of models fitted to response distributions. Third, the group is unnamed at evaluation: items are scored with prompts that never mention a country, so what the adapters

carry sits in the conditioning and not in a country-token channel. We built the construction on a purposive development panel of 16 countries and evaluate it on every country covered by both the GPS and Wave 7 of the WVS, with a second set of adapters for all 64 possible sign profiles on a newly generated scenario bank.

The evaluation has four parts. The first asks whether training installs the synthetic labels on pairs the adapters never saw, whether the adapters' order of countries on the anchor survives a change of scenario bank, and whether a prompt that names the country or one that states the signs does as well. The second is a preregistered screen of 223 Wave 7 items that no adapter was built for: a probe that reads only the adapters' scores assigns each item a dimension and a direction, a fixed rule selects the strongest assignments, and only then are they compared with human country differences. That criterion is retrospective, because the survey existed before the probe. The third is a factorial over the 64 sign profiles, which asks whether the six declared dimensions behave as separate inputs. The fourth, added after the registered results had been read, is a set of descriptive analyses that set the adapters beside the respondents' composites and answer distributions.

The evaluation returns a specific profile. At the first link the method holds in a limited sense. Training installs the synthetic labels on scenarios the adapters never saw, and on trust the adapters keep the GPS order of countries when the scenario bank is replaced, whereas a prompt that names the country does not. A prompt that states the same six signs keeps the trust order as well, and the contrast between the adapters and that prompt is unresolved, so the evidence cannot credit the trained weights over a prompt. The dimensions are entangled: each declared sign moves its own surface in the intended direction and also moves others, often by more, and trust is the one surface that the other signs leave largely intact.

At the second link the evidence changes character. The retrospective screen gives a weak signal: the registered null is rejected, yet the signal is a small fraction of what two human survey waves achieve on the same items, its country-bootstrap interval reaches zero, and registered contrasts do not distinguish the adapters from careful readers of the questionnaire on the few items they share. No dimension passes both the second-bank check and the screen. Descriptively, the adapters' order of countries does not detectably agree with the respondents' composite on any surface, and the GPS's own agreement with those respondents is loose. Two questions that a single agreement figure would merge separate here: whether the instrument keeps the anchor's order of countries, and whether it reproduces the answer distributions of respondents. The adapters keep the order on trust and sit farther from the survey's answer distributions than the base model does, whereas a country-named prompt sits closer and has no trust order. The separation of order from distributional fidelity is known (da Silva et al., 2026; Wang, 2026); the contribution here is a case in which one method, on one anchor, answers the two questions in opposite directions.

Our contributions are three.

1. **An inspectable conditioning method.** Declared aggregate signs, synthetic preference pairs and an unnamed group at evaluation define a construction whose information budget is on the record, so a reader can see what the method was given and check where its readings leave people.

2. **A reusable evaluation protocol.** We separate the three links and state what a favorable or unfavorable result at each permits. It includes a screen of new items whose probe is fixed before any human value is read, a permutation null that absorbs the probe's selection rate, a human ceiling from two survey waves,

item-matched comparators that read only the questionnaire, and a factorial that tests whether declared dimensions stay separate.

3. **A specific empirical failure profile.** The method preserves the labels and the trust order, entangles dimensions, screens new items weakly, and moves away from respondents' answer distributions while a country-named prompt moves toward them.

What the paper shows is bounded by its design: one anchor, one instrument family and one base model. It does not show that trained weights improve on a prompt that states the same signs, that the adapters agree with respondents beyond the weak retrospective signal, or that screening would change which items a researcher fields; pre-piloting is a prospective use that remains untested. The instrument carries no information about magnitudes or about variation among people within a country, and its readings concern differences among countries. Within these bounds a survey designer can consult it beside human pretesting as a complement, never in its place. Section 2 places the method among current uses of synthetic respondents and in the measurement literature on cross-national comparability, Section 3 describes the construction, Section 4 the design and its registration, Section 5 the results, and Section 6 what they permit.

2 Synthetic Respondents as Measurement Instruments

Synthetic respondents and the criterion they need. A language model conditioned on a description of a population can be asked to stand in for a sample of its members. In silicon sampling the description enters the prompt and the model's

answer distribution is read as an estimate of the population's (Argyle et al., 2023); in the homo silicus program the model plays an economic agent in a stylized decision problem (Horton et al., 2023); a systematic review maps how fast these uses have spread across survey research (Buskirk et al., 2026). Fine-tuning on survey responses is one response to the evidence on prompted models (Suh et al., 2025; Cao et al., 2025; Krsteski et al., 2026). Two reviews in this journal draw the conclusion that matters here: validation against human data remains essential (Kozlowski and Evans, 2025), and model predictions are best treated as potentially informative observations while human subjects remain the gold standard (Broska et al., 2025). Both require a criterion against which the synthetic reading can fail. Our evaluation supplies one criterion for each link of the chain.

Construct, measure, and anchor. Three objects have to be kept apart. A construct is a theoretical preference attribute, such as generalized trust. A measure operationalizes it through observed answers, such as the response distribution to a WVS question about whether most people can be trusted. An anchor is an aggregate score that conditions the instrument, such as a country's GPS trust coordinate. The GPS supplies anchors for six economic preferences (trust, risk taking, patience, altruism, positive reciprocity and negative reciprocity), measured with experimentally validated modules in a common design across countries (Falk et al., 2018). We treat these coordinates as declared aggregate moments: the researcher chooses them, and nothing in the construction certifies them as the right moments for a given survey. Mapping a WVS item to a GPS dimension makes the item a candidate indicator, and an average of such items a proposed composite. Agreement between instrument and anchor, between instrument and people, and between anchor and people are three different facts, and they are the three comparisons that the chain makes. The

multitrait–multimethod logic of Campbell and Fiske (1959) applies directly: the human survey is the other method, and an instrument that agrees with its own anchor has passed a construction check without yet meeting a criterion.

Cross-national comparability. The unit of analysis is the country. Country-level agreement does not establish that an item measures the same construct for individuals in different countries, because loadings and intercepts can differ across groups in ways that leave country means comparable for some purposes and misleading for others (Steenkamp and Baumgartner, 1998; Davidov et al., 2014). Work in this journal argues that country-level composites can suffice for simple group comparisons despite noninvariance while understating uncertainty (Bratt, 2025), and that unaddressed measurement error and, less so, noninvariance bias multilevel estimates (Pokropek et al., 2025). Every comparison with respondents below concerns country rankings, with no claim about invariance or individual-level validity.

What is left open. Prior evaluations score models on questions whose human answers are known, or fine-tune them on human survey responses. What remains open is, link by link, what an instrument built from declared aggregate inputs alone preserves of its anchor and of the respondents' answers, with the selection rule for new items registered before the human criterion is read.

3 The Instrument

The instrument is the middle link of the chain. It maps declared aggregate preference signs from the anchor into choice policies and reads them on survey items without naming any country. A language model is treated throughout as a conditional distribution over response sequences given a prompt (Ludwig et al., 2026).

3.1 From declared signs to a shared pair bank

Let $z_c = (z_{c1}, \ldots, z_{cK})$ be the GPS coordinates of country c on the K dimensions trust, risk taking, patience, altruism, positive reciprocity and negative reciprocity. Only the sign profile s_c, with $s_{ck} = 1\{z_{ck} \geq 0\}$, enters the construction. A teacher model and a response generator produce a shared bank of scenarios, each with two responses intended to express a higher and a lower level of one dimension k_i. For country c, the chosen and rejected responses on pair i are

$$(y_{w,ci},\, y_{\ell,ci}) = \begin{cases} (y_i^+,\, y_i^-), & s_{c,k_i} = 1, \\ (y_i^-,\, y_i^+), & s_{c,k_i} = 0. \end{cases} \tag{1}$$

The scenario and response text is identical for every country; only the label changes. No scenario names a country or reuses WVS question wording, and the intended loading of each pair is a generator judgment, not a verified trait manipulation. Two countries with the same sign profile receive identical labels, so the instrument cannot distinguish them, and nothing in the construction carries the magnitude $|z_{ck}|$. That is the information budget: six bits per country.

3.2 Preference optimization as regularized pairwise choice

Each sign profile is fitted with Direct Preference Optimization. Under the Bradley–Terry model, the probability that y_w is chosen over y_ℓ after prompt x is $\sigma\{r(x, y_w) - r(x, y_\ell)\}$, with σ the logistic function and r a latent utility index; this is the binary logit of random-utility theory (Bradley and Terry, 1952). DPO writes the index as $r(x, y) = \beta \log\{\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)\}$ up to a prompt-specific constant that cancels from pairwise differences, and minimizes the negative log likelihood of the labeled

pairs,

$$\mathcal{L}_c(\theta) = -\frac{1}{|\mathcal{D}_c|} \sum_{(x,y_w,y_\ell)\in\mathcal{D}_c} \log \sigma\left\{\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_\ell \mid x)}{\pi_{\text{ref}}(y_\ell \mid x)}\right\}. \tag{2}$$

The parameter β penalizes departures from the reference policy, which makes the estimator a regularized Bradley–Terry fit (Rafailov et al., 2023). Because the labels are deterministic, the likelihood is a fitting criterion and says nothing about noise in human choices. Each adapter is a low-rank update to a quantized Llama 3.1 8B Instruct base model (Hu et al., 2022; Dettmers et al., 2023), trained jointly on all six dimensions for one epoch on 526 of the 658 pairs, with LoRA rank 16 and $\beta = 0.1$. A joint adapter does not isolate any single dimension; whether the dimensions separate in practice is the subject of the factorial in Section 4.

3.3 Two cohorts and two prompt routes

The first cohort has one adapter for each of the 42 countries in the intersection of the GPS and WVS Wave 7, trained on the original scenario bank. These countries realize 26 distinct sign profiles. The second cohort is trained on a second, independently generated bank of 658 pairs under the same sign rule and text restrictions, and it contains one adapter for each of the 64 possible sign profiles, a full 2^6 factorial in the six sign bits. Because the labels depend on the country only through its sign profile, each country takes its profile's adapter in country-level comparisons; most of the 64 profiles are design cells that no country realizes.

The sign-profile prompt route gives the base model the same six signs in words, with no country name and no adapter. It carries the same declared information through a different channel and is the natural comparison for the trained route. The country-named persona route replaces the neutral opening of the evaluation prompt

with "Answer this questionnaire as a typical adult living in" followed by the country's name, and supplies no signs. It changes the information as well as the channel, because the model can draw on whatever it associates with the country name. Naming the population is the comparison a survey researcher would reach first, so we score it on all 42 countries.

3.4 Readout

For an item q with candidate response codes j, the instrument computes the teacher-forced log probability of each code after the rendered item prompt and normalizes over the supplied codes,

$$\widehat{P}_u(j \mid q) = \frac{\exp \ell_{uqj}}{\sum_h \exp \ell_{uqh}}, \qquad y_{uq} = \sum_j g_q(j)\, \widehat{P}_u(j \mid q), \tag{3}$$

where ℓ_{uqj} is the summed token log probability of code j for unit u (an adapter or a prompted profile) and g_q is the item's recode. The score y_{uq} is the expected recoded response. Apart from the persona route, no evaluation prompt names a country. The prompt template, option order and recode are frozen in a scoring specification whose hash every run must reproduce before a model is loaded. On the original 23-item WVS battery, item scores are averaged into seven surfaces: trust, risk taking, altruism, positive reciprocity, negative reciprocity, and two separate patience items. We call this the strict battery. The normalized code probabilities are a scoring convention, and their dispersion need not match any human population's response distributions.

4 Evaluation Design

The evaluation follows the chain, and each comparison permits one kind of statement. A construction check asks whether the adapters encode what they were trained on (anchor to instrument). A second-bank check asks whether the anchor's order survives a change of scenario text. A factorial asks whether the declared dimensions behave as separate inputs. A held-out screen asks whether the instrument's readings of new items agree with an existing human survey (instrument to respondents). Descriptive analyses added after the registered results had been read place the instrument beside the respondents and beside its own labels. Every statistic carries the label of the surface it was computed on (Table 1), and statistics from different surfaces are never pooled.

Table 1: Evaluation surfaces. Every statistic belongs to one, and the status column says how its analysis was fixed.

Label	Units, items and comparison	Status
S1	First-cohort country adapters, strict battery: adapter–GPS rank correlation (model side only)	Basis of the second-bank rule
S2	Second-cohort profile adapters: same statistic as S1	Rule fixed before training
H	Second-cohort adapters on held-out Wave 7 items: signed agreement of selected items with human country means	Preregistered; retrospective criterion
F	Second-cohort adapters and prompted profiles: main effects of six sign bits on seven surfaces (model side only)	V5 planned; rest exploratory
R, N	One unseeded retrain; two near-zero refits (model side only)	Predeclared checks
P	Country-named prompt on the base model: persona–GPS rank correlation (model side only)	One contrast planned; rest descriptive
D	Adapters, factorial and prompt routes: agreement with the GPS and the Wave 7 composite, distance to survey distributions, decomposition of main effects, label recovery	Descriptive, not preregistered

4.1 Second-bank check

For each surface of the strict battery we compute the Spearman correlation between the adapters' surface composites and the GPS coordinate of the surface's own dimension over the 42 countries, first for the first cohort (S1) and then for the second cohort with each country assigned its profile's adapter (S2). Intervals come from a profile-block bootstrap that resamples the 26 realized sign profiles, with 2,000 draws, seed 20261011, and percentile limits. The imprint set is the 4 dimensions whose first-cohort interval excluded zero: trust, risk taking, positive reciprocity and altruism; patience and negative reciprocity were flat and are reported without interpretation. The rule was fixed before the second cohort was trained: for an imprint dimension whose second-bank interval includes zero, no sentence about carry-over to the second bank is written. We also report the agreement between the two banks over the 26 shared profiles; it measures how much the scenario text changes the instrument and involves no human data.

4.2 The country-named prompt

The persona route was scored on the strict battery for each of the 42 countries in one run on the frozen stack. We compute the persona–GPS correlation for each surface (P) and one planned contrast, fixed in a design note committed before any persona was scored: the first cohort's trust correlation minus the persona route's, recomputed on every draw of the profile-block bootstrap above. All other differences are descriptive.

4.3 The held-out screen

Item pool. The pool was built from WVS Wave 7 metadata alone: ordinal questions with two to eleven contiguous response codes outside the battery used to develop the

adapters. A text review excluded items with non-monotone codes or factual content and repaired wording that did not extract, which leaves 223 items that no adapter was built for (online supplement).

Probe. All 64 second-cohort adapters scored every pool item under the frozen specification. For each item i, the probe regresses the 64 adapter scores on an intercept, the six sign bits coded ± 0.5, and their 15 two-way interactions, which leaves 42 error degrees of freedom. With t_d the main-effect t statistic of dimension d, the item's assigned dimension is $d_i = \arg\max_d |t_d|$, or none when the largest $|t_d|$ is below 2.0, and its sign is $s_i = \text{sign}(b_{d_i})$. The primary rule keeps items whose assigned dimension is in the imprint set and whose $|t|$ is at least 2, up to the top 25 percent of all pool items by $|t|$, which caps the selection at 56 items. The secondary rule keeps every imprint item above the threshold and is a reliability check only. The probe reads model scores only; the permutation null absorbs the rate at which noise items cross the threshold.

Statistic and criterion. For each selected item, the signed item agreement is

$$a_i = s_i \cdot \rho\big(\bar{h}_{ci},\, z_{c,d_i}\big), \qquad A = \frac{1}{|S|} \sum_{i \in S} a_i, \tag{4}$$

where ρ is the Spearman correlation over the 42 countries, $\bar{h}_{ci}$ is the survey-weighted Wave 7 mean of item i in country c with negative codes set to missing, and S is the selected set. An item–country cell that was never fielded enters as missing and is not imputed. A is the mean signed agreement with human country differences: it is positive when those differences run in the direction the instrument assigned, against the GPS coordinate of the assigned dimension.

Inference. The registered test is a permutation null over items: draw as many items as were selected from the pool, give them the selected items' dimension and sign pairs in random order, and recompute A; 10,000 draws, seed 20261015, one-sided p equal to the share of null draws at or above the observed A. A 95 percent interval for A comes from 2,000 bootstrap resamples of the 42 countries. The readings were fixed in advance: if A is not above the null at one-sided $\alpha = 0.05$, the adapters do not screen better than random assignment; if A is above the null but below one third of the human ceiling, a screening signal exists but is weak.

Human ceiling. For each selected item fielded in both WVS waves 6 and 7 we correlate the two waves' country means over the 28 countries with both waves (World Values Survey Association, 2022) and average over items, which gives the agreement two human surveys reach on these items; the ratio of A to it is a benchmark only.

Per-dimension readings. The registered analysis also reports A_d, the mean of a_i over the selected items assigned to dimension d, with an interval from the same country resamples. These intervals are unadjusted, so we add exploratory Bonferroni intervals for the four imprint dimensions from the same draws at the 98.75 percent level.

Comparators. Two comparators read only the questionnaire and saw no model score or human response: a face-valid rule that assigns an item to the one dimension whose construct words its wording contains, and a typed decision model (jev-1.13.0) run 5 times, which assigns an item when at least 3 runs agree on dimension and sign (online supplement). The registered primary contrast is coverage-matched, $D_{\text{common}} = A_{\text{adapter}} - A_{\text{comparator}}$ on the items both assign, with an interval from the paired country bootstrap. An interval above zero would mean the adapters find

structure the wording does not reveal; one that covers zero was registered as "the adapters match a careful reader"; one below zero would favor the reader. The contrast on each route's own items is reported beside it, and the preregistration named in advance that both comparators concentrate on trust.

Readout variants and the prompt route. The probe was re-estimated on three re-scorings of the pool (options reversed, two prompt paraphrases), and the selection was declared dependent on wording if more than half of the selected items changed dimension or sign. The same probe and test were applied to the 64 prompted profiles.

4.4 Registration, blinding and amendments

The screen was registered in a document committed to the project repository and timestamped by a pushed git tag before any held-out item was scored by an adapter, when no selection existed and no held-out human value had been read. A freeze guard enforced the order of work (a timeline is in the online supplement): score the held-out items, run the probe, freeze the hashes of the selection and the item pool, and only then join the human data.

Two amendments were made, both before any human held-out value was read. Amendment 1 repaired 12 items whose standalone wording could not be answered as written, under the pool rule and override mechanism already registered, and every route was re-scored as one run. We disclose that the first probe pass had been computed and read before the repairs were chosen, and that two repaired items (Q153 and Q157) were in its top quartile. The comparators had assigned the pool on the pre-repair wording, so Amendment 2 removed the repaired items from both comparator routes before any comparator statistic was computed; it affected one item (Q82), not in the

adapter selection. That decision preceded any human value; the amendment text, written after the primary statistic was read, records it.

The criterion is retrospective. The Wave 7 file is public, existed before the probe, the pool and the adapters, and is the only human source in this study. Registration guards against tuning the probe to the criterion; it cannot make the criterion prospective, so results on surface H are labelled retrospective throughout. The online supplement discloses the provenance gaps, none of which touches a statistic reported here.

4.5 The sign-profile factorial

The second cohort covers all 64 sign profiles, so the strict-battery composites form a full factorial in six binary factors. For each of the seven surfaces we fit the probe's regression. A main effect is the difference between the mean composites of profiles with the bit set and without it. Because the bits were assigned by design, this is a causal statement about the instrument and carries no implication about how preferences operate in people. Each bit has an own surface (patience owns two). The analysis plan set one acceptance check, separate from the held-out registration: at least 4 of the six bits must move their own surface more than any other (V5), against a binomial reference with a one-in-seven chance of a hit. Exploratory views divide effects by each surface's standard deviation over the 64 profiles, count the surfaces whose largest mover is their own bit, and report the first principal component's variance share and the mean absolute correlation between surfaces. The same analyses on the 64 prompted profiles contrast two instruments that carry the same declared information.

4.6 Retraining noise and the near-zero margin

One profile adapter was retrained without a fixed seed, which gives a noise floor for the pipeline. A predeclared sensitivity check also removes the training pairs of every dimension whose GPS coordinate lies within 0.10 of zero and refits two countries (Great Britain and Colombia), each beside a size-matched control; the rule was to report the margin and never to promote it.

4.7 Descriptive follow-up analyses

Four further analyses were added after the registered results had been read: three columns of agreement, distance to the survey distributions, a decomposition of the factorial's cross effects, and held-out label recovery. The last two are defined with their results. None is preregistered or tests a rule fixed in advance; we label their numbers D and read them as descriptions. Their human input is the Wave 7 file used for the screen, so every comparison with respondents is retrospective.

Three columns. For each of the seven surfaces we compute three Spearman correlations over countries: the adapter composite with the GPS coordinate of the surface's own dimension, the adapter composite with the Wave 7 composite of the same items, and that human composite with the GPS. The adapter columns use the first-cohort adapters (the second cohort is in the supplement) with the profile-block bootstrap above, and the human–GPS interval resamples countries (seed 20260905). The human columns of risk taking cover one country fewer, which has no valid answer to the risk item.

Distance to the survey distributions. For country c and item q, the total variation distance between the model's option probabilities and the survey-weighted

human answer shares is

$$\mathrm{TVD}_{cq} = \tfrac{1}{2} \sum_{j} \left| \widehat{P}_c(j \mid x_q) - P_c^H(j \mid q) \right|,$$

where j runs over the answer codes and P_c^H is the share of valid human answers equal to j. The headline statistic is the mean over items within a country and then over countries. We score four cells on the same frozen stack: the base model, the first-cohort adapters, the country-named persona prompt on the base model, and the adapters with the persona prompt together (gates in the supplement). Contrasts are paired differences of country means, each with a profile-block and a country interval (2,000 draws each). The interaction contrast is the adapters' effect with the persona prompt minus their effect without it; zero means the two add. The human answer shares enter the distances and are not reported.

5 Results

The results follow the chain, and each statistic belongs to one surface of Table 1. Sections 5.1 to 5.3 concern the link from anchor to instrument: what training installs, which order of countries survives a new scenario bank and what carries it, and whether the declared dimensions stay separate. Sections 5.4 to 5.7 concern the link from instrument to respondents: the preregistered screen and its comparators, agreement with the respondents' composites, and distance to their answer distributions. Everything computed against the Wave 7 file is retrospective. Four figures carry the evidence beside the text that reads them, and exact values are in the text, Table 2 and the online supplement.

5.1 Anchor to instrument: what training installs

Every training pair of a country carries a label: the completion that goes with the sign of the country's GPS coordinate on the pair's own dimension is the chosen one. The first question is whether that installation carries over to pairs the adapters never saw. This is a construction check on the synthetic labels (descriptive, no human data). Training pairs were split once into a training part and a held-out part that the adapters never saw. For each of the 42 first-cohort adapters, the margin is the difference between the log probability of the chosen and of the rejected completion, scored under the adapter and under the base model in a further run on the frozen stack (gates in the online supplement). Recovery is the share of held-out pairs on which the adapter's margin exceeds the base model's, accuracy is the share on which the adapter prefers its own chosen completion outright, and base accuracy is the same share for the base model. We report country-equal means with a profile-block bootstrap.

Recovery is 0.940 [0.926, 0.952] and accuracy is 0.816 [0.771, 0.856]. The base model's accuracy is 0.489 [0.471, 0.506], an interval that includes one half, so the base model shows no detectable preference for the chosen completions before training. The mean margin shift is 40.3 [28.2, 54.1] nats. Training installs each country's synthetic labels on scenarios the adapter never saw, which is construction fidelity. It says nothing about whether the labels order countries like the GPS and nothing about respondents, and the distance results below show that the same adapters lie farther from the survey's answer distributions than the base model does.

5.2 Anchor to instrument: the trust order and what carries it

The first cohort encodes four of the six declared signs on the strict battery (S1). Over the 42 countries, the adapter–GPS Spearman correlation is 0.744 [0.500, 0.865] for trust, 0.435 [0.085, 0.686] for risk taking, 0.488 [0.120, 0.695] for positive reciprocity, and 0.406 [0.020, 0.676] for altruism. These four form the imprint set that the screen later uses.

The second cohort tests whether the order survives a change of scenario text (S2). Trust keeps its agreement with the GPS, at 0.803 [0.615, 0.876]. The other three imprint dimensions keep positive point estimates, but their intervals now include zero (Table 2), so 1 of the 4 imprint dimensions passes the second-bank check and 3 do not. Under the rule fixed before the second cohort was trained, we write no sentence about the other three carrying over. Trust is the dimension on which retention of the anchor's order is replicated across scenario banks. The banks also agree profile by profile: over the 26 shared profiles the rank correlation between first-bank and second-bank trust composites is 0.907 [0.774, 0.957], and risk taking is the least stable surface, at 0.544 [0.187, 0.772], which fits its loss on the second bank (the online supplement gives every surface). This agreement measures how much the scenario text changes the instrument and says nothing about people.

What carries the trust order? The comparison a survey researcher would reach for first is a prompt that names the population, and it does not reproduce the order (P): the persona–GPS correlation over the 42 countries is 0.028 [−0.339, 0.381]. The planned contrast with the first cohort is 0.716 [0.191, 1.154], and the reading is the same under a country bootstrap. A prompt that states the same six signs in words, with no adapter and no country name, tracks trust as closely as the adapters do, at

0.773 [0.584, 0.844] (panel A of Figure 4, which sets all routes side by side). On trust, what separates the routes is whether the declared sign is supplied, whichever channel carries it. The contrast sets declared signs against a country name and says nothing about trained weights against prompting. On the other surfaces the persona route's intervals include zero, except for patience, which it orders positively on one item and negatively on the other (online supplement).

5.3 Inside the instrument: the sign-profile factorial

The factorial asks whether the six declared signs behave as separate inputs (F). On the trained route, every sign bit moves its own surface in the intended direction (panel A of Figure 1; the online supplement gives each effect with its t statistic). All seven own effects are positive, from 0.364 for trust ($t = 16.74$) to 0.124 for positive reciprocity ($t = 3.02$), and 0 of 7 point the wrong way.

The planned separability check fails. Only 2 of the 6 bits, those for altruism and negative reciprocity, move their own surface more than any other surface, where at least 4 were required (binomial $p = 0.207$). Scale-free views agree (online supplement). Each bit moves its own surface and also moves others, often by more: the risk-taking bit moves the altruism surface by 0.983, against the altruism bit's own effect of 0.242, and the trust bit is the largest mover on 3 of the 7 surfaces. One common direction carries 51.3% of the variance of the seven standardized surfaces, and the mean absolute correlation between surfaces is 0.395. The sign bits are inputs that the instrument keeps distinct in direction and mixes in magnitude.

The prompted route fails the check by the same count, 2 of 6 (binomial $p = 0.207$), so V5 does not separate the routes; the common direction and the sign of the own effects do. Under prompting the first principal component carries 83.5% of the variance,

the mean absolute correlation between surfaces is 0.807, and the trust bit is the largest mover on all 7 surfaces. Three prompted own effects point against the intended direction: risk taking at -0.457 ($t = -22.31$), patience Q43 at -0.069 ($t = -19.75$), and negative reciprocity at -0.023, which at $t = -1.80$ is not distinguishable from zero. These are behavioral cross effects of the instrument. The contrast concerns entanglement and the direction of own effects and is silent on screening accuracy, where the difference between routes is unresolved.

Why does the trust order survive when the other surfaces lose theirs? We split the second-cohort adapters' main effects into the effect of each surface's own bit and the summed effect of the other five, evaluate each part at the sign profile of each of the 42 countries, and correlate it with the GPS coordinate of the surface's own dimension (D; the online supplement gives every surface). This is a behavioral decomposition that uses model quantities and published GPS values only, and it says nothing about mechanism inside the network. The own-bit part orders the countries like the GPS on all seven surfaces, with correlations from 0.817 to 0.866. That ordering is mechanical: for a positive own effect, as all seven are, the own part is proportional to the sign of the GPS coordinate, so it restates the correlation between signs and GPS and is no separate test of learning. The full prediction, which adds the cross bits, keeps the GPS order on trust, 0.834, and loses most of it elsewhere, with correlations from -0.020 for patience Q43 to 0.422 for altruism. The cross part carries little of the full prediction's variation on trust and most of it elsewhere: Var(cross)/Var(full) is 0.147 on trust and runs from 0.830 to 1.049 on the other six. This is a ratio of variances and no percentage of variance explained; it exceeds one on patience Q43 because the own and cross parts covary negatively. The comparison sets trust against six surfaces, so it is no graded relation and we claim no dose–response. No bank label depends on a second dimension: in both banks all 27,636 country-pair training rows follow the

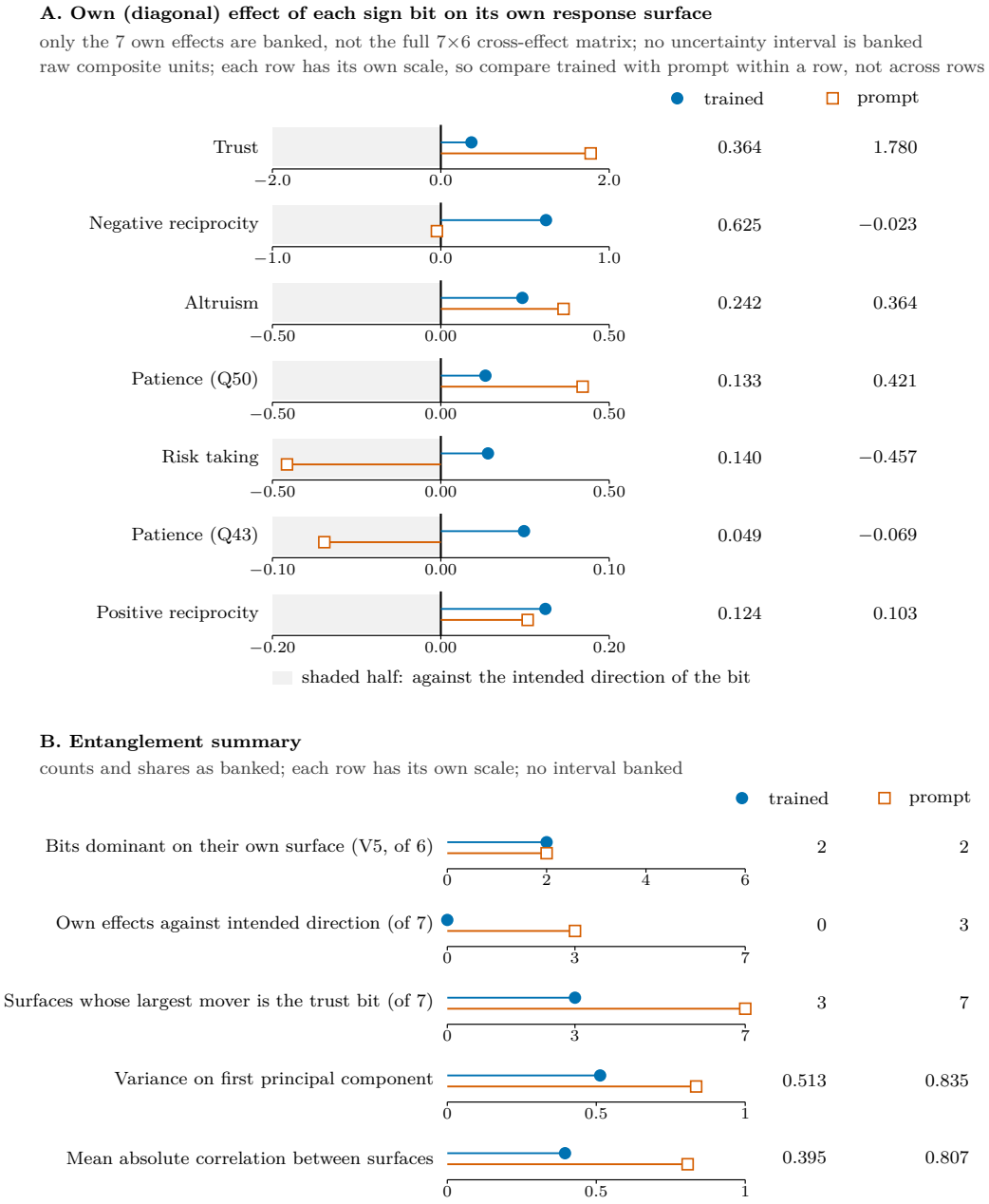


Figure 1: Own effects of the sign bits in the sixty-four-profile factorial (surface F; model side only). (A) For each of seven strict-battery surfaces, the main effect of the surface's own sign bit, for the trained second-cohort adapters (circles) and the sign-profile prompt (squares). All 7 trained own point estimates have the intended sign, and 3 prompted ones do not. (B) Counts and shares that summarize entanglement. V5 was a planned check; the other rows are exploratory.

sign of the pair's own dimension, with 0 mismatches in the first bank and 0 in the second, so the labels do not carry the cross talk. We claim no cause of the loss; that would need component-level analysis and interventions.

Retraining noise is small beside these effects: an unseeded retrain of one profile moves every strict surface by at most 0.112 of a standard deviation across profiles (R; online supplement). This is a narrow observed comparison of one retrained unit with its original; variation across units was not measured.[1]

5.4 Instrument to respondents: the retrospective screen

Retrospectively, the screen rejects its registered null (H). The probe selected 56 of the 223 pool items under the primary rule. Their mean signed agreement with the Wave 7 country differences is $A = 0.061$ (Figure 2, panel A), with a country-bootstrap interval of $[-0.007, 0.126]$. Under the permutation null, which reassigns the same dimension and sign pairs to items drawn at random from the pool, the null distribution has mean 0.012 and ninety-fifth percentile 0.046, and the one-sided p is 0.010 (panel B).

By the registered reading, the signal is weak. Two WVS waves agree with each other on the selected items fielded in both waves with a mean Spearman correlation of 0.746 (median 0.767, 34 items). The screen's A is 8.1% of that ceiling, well below the registered bar of one third, which would be 0.249. The interval reaches zero while the permutation test rejects the null because they answer different questions: whether the assignments beat random assignments of the same pairs to items from the same pool, and how much A varies across samples of countries. The direction of the finding is clearer than its size. The registered reliability checks point the same way. The secondary rule gives $A = 0.070$ over 185 items with a one-sided permutation $p < 0.001$,

[1]The near-zero filter (N) is not promoted under the predeclared rule, and two refits support no claim of a reversal across countries; the online supplement gives the margin in full.

and dropping one country at a time moves the primary A only between 0.051 and 0.072.

The per-dimension readings show where the signal sits (Table 2). Under the primary rule, only the items assigned to risk taking have an interval above zero, $A_d = 0.158$ [0.015, 0.290] on 13 items, of which 92.3% agree in sign. The intervals for trust and altruism include zero. The items assigned to positive reciprocity point the wrong way: their point estimate is negative, and only 18.2% of the 11 items agree in sign. Under the exploratory Bonferroni adjustment for four dimensions, the interval for risk taking becomes $[-0.026, 0.326]$ and includes zero.[2]

The two checks disagree by dimension. Trust survives the second bank, and its screen interval includes zero. Risk taking is the one dimension whose screen interval excludes zero before adjustment, and it is one of the three that lose their interval on the second bank. No dimension passes both registered checks, and the per-dimension cells are small.

5.5 Instrument to respondents: questionnaire readers and the prompt route

The registered contrasts do not distinguish the adapters from a careful reader of the questionnaire (Figure 2, panel C; the online supplement tabulates every contrast). Every interval covers zero, which the preregistration reads as "the adapters match a careful reader". That reading rests on few items: the coverage-matched contrasts use the 5 items that the adapters and the face-valid rule both assign and the 9 items they share with the typed decision model, and the routes agree on dimension and sign for 1

[2]The probe's per-dimension permutation p values were not registered. For altruism under the primary rule they point the other way from the registered interval, so we report the registered intervals only.

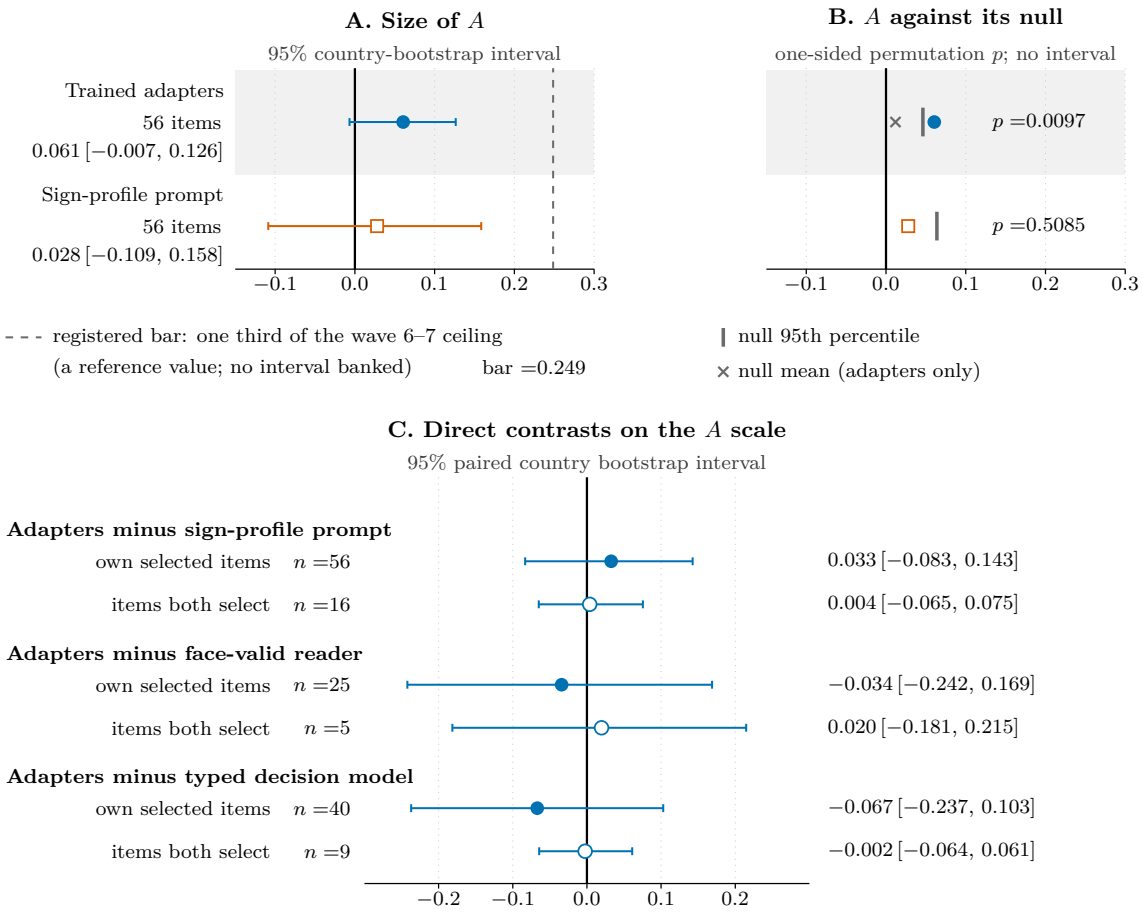


Figure 2: The retrospective held-out screen (surface H; preregistered, retrospective criterion). A is the mean signed agreement, over the selected items, between the direction the probe assigned to an item and the WVS Wave 7 country differences on that item. (A) A for the 56 items the adapters selected and for the sign-profile prompt's own selection, with 95% country-bootstrap intervals over 42 countries; the dashed line is the registered bar of one third of the two-wave human ceiling, a reference value with no interval. (B) The same A against its permutation null: the bar is the null's ninety-fifth percentile, the cross its mean (adapters only; none is banked for the prompt), and the one-sided p is printed. (C) Paired contrasts, adapters minus each comparator, on each route's own items and on items both select (n printed), with 95% paired country-bootstrap intervals; the registered primary contrasts are those with questionnaire readers on shared items. Interval and permutation p answer different questions, and an interval covering zero is not evidence of equivalence.

Table 2: The second-bank check and the retrospective screen, by imprint dimension. The two checks are reported side by side and never averaged.

Dimension	S2: second bank ρ [95% interval]	H: primary rule A_d [95% interval], items	H: primary rule, Bonferroni A_d [adjusted interval]	H: secondary rule A_d [95% interval], items
Trust	0.803 [0.615, 0.876]	0.062 [−0.047, 0.173], 12	[−0.088, 0.199]	0.092 [−0.019, 0.199], 44
Risk taking	0.210 [−0.199, 0.584]	0.158 [0.015, 0.290], 13	[−0.026, 0.326]	0.143 [0.045, 0.229], 50
Positive reciprocity	0.366 [−0.068, 0.652]	−0.071 [−0.281, 0.154], 11	[−0.332, 0.205]	−0.049 [−0.198, 0.117], 22
Altruism	0.364 [−0.057, 0.653]	0.069 [−0.036, 0.177], 20	[−0.058, 0.198]	0.040 [−0.038, 0.117], 69

Note: S2 is the Spearman correlation between second-bank adapter composites and the GPS coordinate over 42 countries, with a profile-block bootstrap (2,000 draws, seed 20261011); model side only. H is the retrospective screen against WVS Wave 7 country means: A_d is the mean signed item agreement over the selected items assigned to the dimension, with the registered country bootstrap (2,000 resamples, seed 20261015). The registered intervals are unadjusted; the Bonferroni intervals over the 4 imprint dimensions are exploratory and read off the same draws. The secondary rule is a registered reliability check. No dimension has an interval above zero on both S2 and H.

and 7 of them. An interval that covers zero on so few items is consistent with a match and does not establish one. Both readers concentrate on trust, where their own-item A exceeds the 0.062 of the adapters' own trust items, but their per-dimension cells are too small to be informative.

Prompting the base model with the same six signs does not pass the registered null. The same probe applied to the 64 prompted profiles gives $A = 0.028$ [-0.109, 0.158] with a one-sided permutation p of 0.509, and it assigns its selected items to a single dimension, trust. The adapter route passes the null and the prompt route does not, but the direct difference between them is unresolved: adapter minus prompt is 0.033 on each route's own items and 0.004 on the 16 items both select, and both intervals cover zero. The two routes agree on dimension and sign for 6 of those shared items. The paper therefore cannot credit the trained weights over the prompt on the screen.

Item-level assignment is stable under option order and paraphrase, whereas membership in the particular selected set churns. The registered statistic, the number of selected items whose dimension and sign are unchanged, is 39 of 56 with reversed options, 39 with the first paraphrase and 30 with the second. No variant changes more than half, so the registered condition for wording dependence is not met, although the second paraphrase comes close. Membership is another matter: the primary rule is a top-quartile cut on $|t|$, and only 18, 29 and 16 of the 56 items are selected under both the original readout and each variant. Dimension and direction are more stable than the list of top-ranked items.

5.6 Instrument to respondents: agreement with the human composite

The next two subsections report descriptive follow-up analyses (D), added after the registered results had been read. None is preregistered and none enters a registered family. They set the instrument beside the respondents of the Wave 7 file on the strict battery, and every statement is retrospective. Figure 3 gives three Spearman correlations over the 42 countries for each of seven surfaces: adapter composite with the GPS, adapter composite with the human composite of the same items, and the human composite with the GPS. The panels have different statistical targets and bootstrap units, so their widths are not comparable.

On trust, the first-cohort adapters track the GPS, 0.744 [0.500, 0.865], and do not detectably reach the human composite, 0.295 [−0.044, 0.547]. All seven adapter–human intervals include zero, which is not evidence that no association exists. The largest point estimate belongs to negative reciprocity, 0.389 [−0.008, 0.694], a surface on which the adapters do not track the GPS. The second-cohort adapters give the same picture on trust, with an adapter–human correlation of 0.340 [−0.025, 0.586]; positive reciprocity is the one second-cohort interval that excludes zero, and only just (online supplement).

The link between the GPS and the human composite is itself loose on these items. Its interval excludes zero on four of the seven surfaces: positive for trust, 0.351 [0.088, 0.564], and negative for patience Q43, −0.533 [−0.741, −0.266]. We read this column as a reference for scale. It is neither an upper bound on what an adapter could reach, because it does not bound a scorer that is nonlinear or constant within sign profiles, nor evidence that the anchor is the bottleneck. Together, the three panels say that the adapters carry the GPS order on trust, that the GPS order is only loosely related

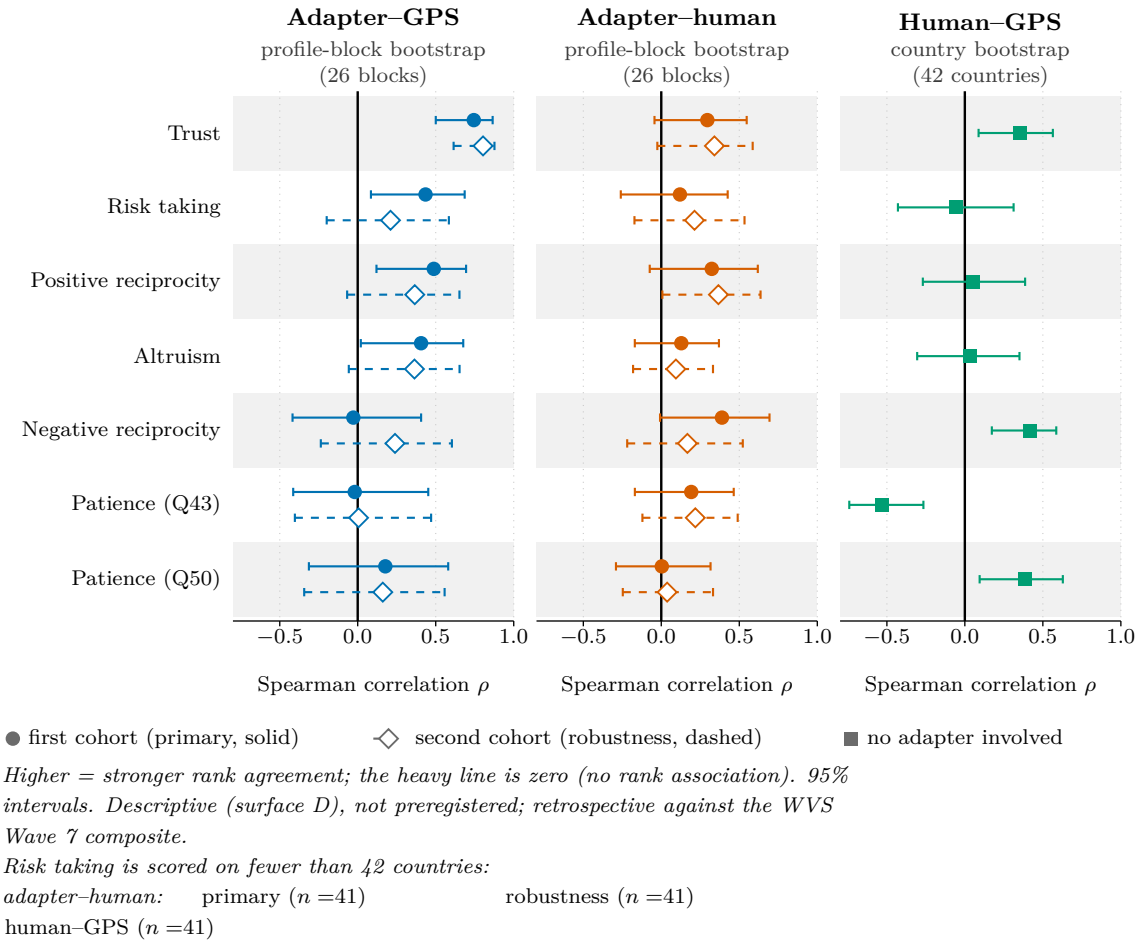


Figure 3: The three links on seven strict-battery surfaces (surface D; descriptive, not preregistered; retrospective). Each mark is a Spearman correlation across countries with a 95% interval; the heavy line is zero. Left: first-cohort adapter composite with the GPS coordinate of the surface's own dimension. Middle: adapter composite with the Wave 7 composite of the same items. Right: that human composite with the GPS. Circles are first-cohort adapters (primary), diamonds second-cohort adapters (robustness), squares involve no adapter. Intervals in the adapter panels come from a bootstrap over the 26 realized sign profiles, those in the right panel from a bootstrap over the 42 countries. Risk taking covers 41 countries in the two human columns.

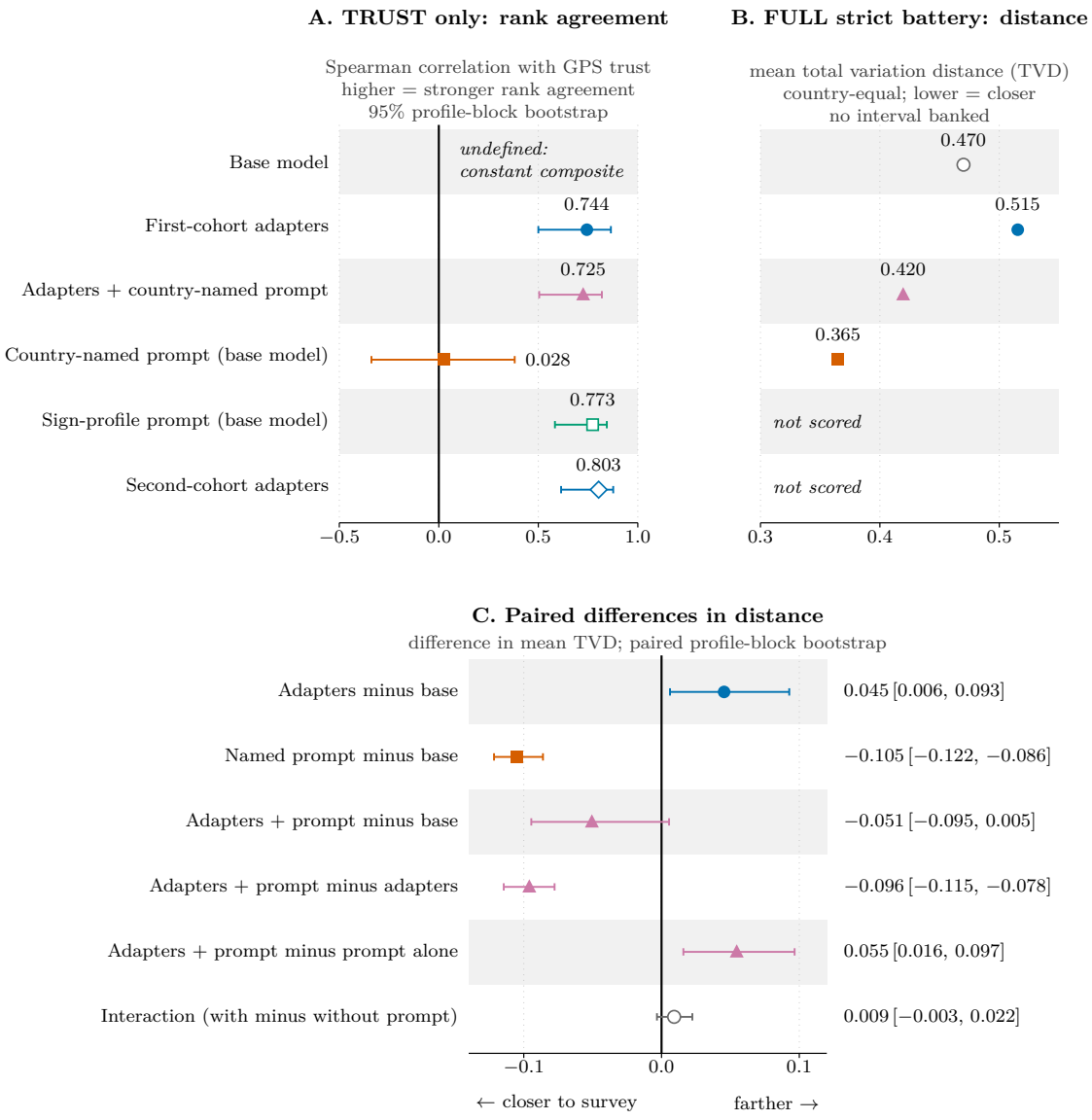
to the human composite, and that the adapters' order of countries does not detectably agree with the human composite on any surface. The exact values for every cell are in the online supplement.

5.7 Instrument to respondents: order versus distance

The correlations above concern the order of countries. A different question is how far the instrument's answer distribution lies from the survey's, measured by the country-equal mean total variation distance (TVD; lower is closer; D) in four cells: the base model, the first-cohort adapters, the country-named persona prompt, and the adapters and the persona prompt together.

The base model's mean distance is 0.470. The adapters sit farther from the survey, at 0.515, and the persona prompt sits closer, at 0.365 (panel B of Figure 4; a cell mean carries no interval, and the paired contrasts of panel C do). Relative to the base model, the adapters are farther by 0.045 [0.006, 0.093] under profile blocks, and the persona prompt is closer by −0.105 [−0.122, −0.086]; country resampling gives the same readings. By dimension, the adapters are farther than the base model on four of the six dimensions and slightly closer on the other two (point estimates; online supplement). With both interventions the distance is 0.420. Adding the persona prompt to the adapters moves the distance by −0.096 [−0.115, −0.078], and adding the adapters to the persona prompt moves it by 0.055 [0.016, 0.097]. The interaction contrast, the adapters' effect with the persona prompt minus their effect without it, is 0.009 [−0.003, 0.022] under profile blocks, which includes zero, so the data show no detectable departure from additivity.

The ordering of the cells by distance does not follow their ordering by trust agreement with the GPS. The persona prompt has the lowest distance and no trust



Descriptive (surface D), not preregistered. The base model's trust–GPS correlation is undefined (constant composite), not zero. Trust is one dimension; TVD scores the full strict battery. The crossing of the two readouts is descriptive: not a causal effect of fine-tuning, nor a general law about weights versus prompts.

Figure 4: Order and distance for the same routes (a descriptive display, not preregistered; the distances are retrospective). (A) Trust only: Spearman correlation of each route's trust composite with the GPS trust coordinate over 42 countries, with 95% profile-block intervals. The first-cohort, second-bank and country-named values are those of Section 5.2, where their own status is stated. (B) Mean total variation distance (TVD) to the survey answer shares over the full strict battery, country-equal; routes not scored on this battery are marked. (C) Paired differences in mean TVD with 95% profile-block intervals; negative means the first-named cell is closer. Panel A scores trust only and panels B and C all dimensions, so they are different metrics.

order (Section 5.2). The adapters have the highest distance and keep the trust order, and adding the persona prompt to the adapters keeps the trust order, 0.725 [0.505, 0.820], while the distance falls. The two measures ask different questions, and neither says why the adapters' distributions lie where they do. The instrument carries no information about magnitudes (Section 6), so a distance to survey distributions is not a quantity it was built to reduce.

6 Discussion

The opening question was what a retained benchmark warrants. The answer is graded. It warrants a statement about the benchmark link: the synthetic labels are installed, and on trust the order of countries survives a change of scenario text. It does not carry to the respondent link, where agreement is weak or undetected and the instrument moves away from the survey's answer distributions.

The conditioning method: what the signs carry. The construction makes its information budget explicit, and that makes the results interpretable. The adapters keep the trust order across scenario banks (Section 5.2). A country name does not recover that order, whereas a prompt that states the six signs does, with a correlation with the GPS on trust of 0.773 [0.584, 0.844] against 0.028 [−0.339, 0.381] for the named country, and the contrast between the adapters and the sign prompt in the screen is unresolved. What carries the trust order is therefore the declared sign, whichever channel supplies it, and the evidence cannot credit the trained weights over a prompt. The statement concerns trust only. The decomposition describes where trust differs. Every own effect is positive, so the own part of the main-effects prediction restates the correlation between the signs and the GPS on every surface. What differs

is how much the other five signs add, which is little on trust, with Var(cross)/Var(full) at 0.147, and most of the variation on the other six, with ratios from 0.830 to 1.049. The decomposition describes behavior and locates nothing inside the network.

Entanglement is the second part of the method's profile. Each sign moves its own surface in the intended direction and also moves other surfaces, one common direction carries about half of the variance across surfaces, and the planned separability check fails on both routes. Prompting with the same signs is more entangled still, and three of its own effects point the wrong way. Training on sign-labeled pairs keeps the direction of each declared input and does not deliver separate dimensions. Entanglement under intervention has been reported for activation steering (Dang and Masud, 2026); the factorial adds a joint measurement for trained adapters and a prompt that carry the same declared information, and makes no claim about the best training recipe.

The evaluation protocol: what it separates. The second-bank check turned four dimensions with a first-cohort signal into one. The screen's permutation test and its bootstrap interval answer different questions, and reporting both made the size of the signal clearer than either would. Coverage-matched comparators showed that the paired contrasts between the adapters and two readers of the questionnaire are unresolved on the few items they share, and the small overlap is part of the finding. The separation of the three links showed that the GPS's own agreement with the human composite is loose on these items, so weak adapter–human agreement cannot be attributed to the instrument alone: it could arise from the instrument, from the mapping between items and the GPS construct, or from limited coherence between the two surveys, and the design cannot separate these. The registered screen and freeze sequence is fixed before the human criterion is read, and the protocol can be

applied to another conditioning method, anchor or criterion; the descriptive analyses added afterwards are labelled as such.

The failure profile: from instrument to respondents. The screen asked whether a probe that reads only the instrument picks items whose human country differences run in the assigned direction, and the registered answer is yes, weakly (Section 5.4). The signal is a small fraction of the two-wave human ceiling and concentrated in risk taking, which the second bank does not support; trust, which the second bank supports, does not stand apart from zero in the screen. No dimension passes both checks, and the adapters' order of countries does not detectably agree with the respondents' composite on any surface (Section 5.6).

Order and distribution. The two questions separated in the introduction receive different answers (Section 5.7). The adapters keep the anchor's order of countries on trust and lie farther from the survey's answer distributions than the base model, by 0.045 [0.006, 0.093] in total variation distance. A country-named prompt lies closer, by -0.105 [-0.122, -0.086], and carries no trust order. If agreement were one quantity, the routes would fall in one ranking, and they fall in two. We offer one reading without testing it: the training signal states which of two responses is preferred and says nothing about how strongly, or about how widely respondents spread across the options. The order of countries is a quantity that signs speak to, whereas a distance to the survey's answer distributions is one they carry no information about. The contrast describes one anchor, one instrument family and one survey, and does not show that the country-named prompt matches respondents. A country name has been found to recover a country ranking in the European Social Survey while supporting no distributional inference (Wang, 2026). The separation of group-level order from

distributional fidelity is established (da Silva et al., 2026); the present evidence adds a case in which the two cross.

What a prospective validation would have to establish. Whether consulting the instrument helps a researcher before fielding is not shown here, and the retrospective design cannot show it. A prospective validation would need three elements. The probe's selections would be made, and a human-only rule for choosing items declared, before any new items are fielded. The selected items would then have to separate countries in the assigned direction in new data, and to do so better than the human-only rule on matched countries and items. And the dimension on which the signal appears would have to be one that the second bank also supports. A result at that level would license the use; a result at the level reported here does not.

Limits. Five limits apply to every result above. First, the human criterion is retrospective and the only human source; registration and the freeze guard protect against tuning the probe to it, but cannot make it independent of the survey program from which the item pool was drawn. Second, the instrument is built from signs and cannot represent magnitudes or differences among countries that share a sign profile or among people within a country, and country-level agreement does not establish individual-level validity or measurement invariance (Davidov et al., 2014; Bratt, 2025). Third, every adapter shares one base model, one training recipe and one readout convention, whose code probabilities have a dispersion that need not match human response distributions. Fourth, the per-dimension cells are small and the selected set churns under readout variants. Fifth, the noise checks are narrow: one unseeded retrain is a narrow observed comparison for one unit, and two near-zero refits support no claim across countries.

Intended use. Within those limits, the instrument has a modest and inspectable use. A survey designer can read the dimension and direction it assigns to candidate items as one input beside expert review and human pretesting, keeping in view that no dimension passed both checks. Its value lies in what it makes explicit: the declared moments, the sign rule and the absence of any group name at evaluation are on the record, so a reader can see what information the instrument was given and check where its readings depart from people. It is a complement to fielding and pretesting, and gives no reason to replace them. In the terms of Broska et al. (2025), its readings are potentially informative observations that still require a human criterion.

7 Conclusion

Preserving an external benchmark is evidence about the link between benchmark and method. For synthetic cultural agents, which compile declared aggregate preference signs into choice policies that read survey items without naming a group, we tested each link of the chain from anchor to instrument to respondents separately.

Between anchor and instrument the method holds in a limited sense. Training installs the synthetic labels on pairs the adapters never saw, and on trust the adapters keep the anchor's order of countries across a change of scenario bank. A country name does not keep that order, and a prompt stating the same signs does, so the declared signs carry the trust order and the evidence cannot credit the trained weights over a prompt. The adapters do not keep the declared dimensions separate. Between instrument and respondents the evidence is weak. A preregistered screen of held-out items, read against a retrospective criterion, gives a small signal that the registered comparisons do not separate from careful readers of the questionnaire, no dimension passes both the second-bank check and the screen, and the adapters' order of countries

does not detectably agree with the respondents' composite. Keeping the anchor's order of countries and approaching respondents' answer distributions are different questions, and on a descriptive reading the adapters keep the first and lie farther from the survey's answer distributions than the base model, whereas a country-named prompt lies closer with no trust order.

The method is an inspectable, group-level reading of candidate items that a researcher may use beside human pretesting, as a complement and never in its place. The evaluation protocol applies to other conditioning methods and other anchors. The test that would license more is prospective: probe-selected items fielded to people after the selection and a human-only comparison rule have both been declared. Until that test is run, pre-piloting remains untested.

Preregistration Statement

The held-out screen (Sections 4.3 and 5.4) was preregistered. The registration document was committed to the project repository and timestamped by the git tag `prereg-heldout-2026-10-14`, pushed before any held-out item was scored by an adapter. Amendment 1 (repair of 12 items whose standalone wording could not be answered) was made after the first model-side probe pass had been computed and read, and before any human value was read. The freeze file was committed, and Amendment 2 (removal of those items from both comparator routes) was decided, before the single read of held-out human data; the text of Amendment 2 was written afterwards and records that earlier decision. The criterion data, WVS Wave 7, are public and existed before the study, so the registration fixes the analysis before the authors read the human values and the criterion remains retrospective. All registered analyses of the screen are reported in Section 5.4, Table 2 and Figure 2, except

two descriptives (item-level agreement after dropping items with small $|a_i|$, and the face-valid reader's statistic on all of its items including negative reciprocity), which are reported in the online supplement. The Bonferroni intervals, the wave-specific screen statistics, the scale-free factorial views and the factorial route contrast are exploratory. The factorial's separability check (V5) was fixed in the project's analysis plan, separately from the registration. The second-bank decision rule was fixed before the second cohort was trained and was not part of the registration. An anonymized copy of the registration is provided with the submission.

Code, Materials, and Data Availability

Analysis code, the item pool and its overrides, the frozen selection, the comparator files, all model scores, and the scripts that build every number in this article from those scores are available for review at an anonymized link. Model adapters and score archives are released at acceptance. The frozen inputs are identified by SHA-256 hashes: the amended item pool `19241f6f103c51d6`... as written to disk, whose line-ending-normalized form in the repository is `a1dbceea1a33da05`...; the overrides file `0f5cd2bbd88ef6ea`...; and the frozen selection `782b7d2123f9758e`.... The pool is regenerated byte for byte from the overrides file by the pool-building script; full hashes and the line-ending convention are in the online supplement. WVS and GPS data are available from their providers (Haerpfer et al., 2022; World Values Survey Association, 2022; Falk et al., 2018) and are not redistributed; no respondent-level data or country-level human means are included in the replication materials.

Institutional Review

This study analyzes publicly available, de-identified secondary survey data and involves no interaction with human participants.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Generative AI Use

Large language models are the object of study in this article. In addition, an AI coding and writing assistant was used under the authors' direction to write analysis code and to draft and edit manuscript text; the authors reviewed all code, numbers and text and take responsibility for them.

References

Argyle, L. P., E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis 31*(3), 337–351.

Bisbee, J., J. D. Clinton, C. Dorff, B. Kenkel, and J. M. Larson (2024). Synthetic replacements for human survey data? the perils of large language models. *Political Analysis 32*(4), 401–416.

Boelaert, J., S. Coavoux, É. Ollion, I. Petev, and P. Präg (2025). Machine bias. how do generative language models answer opinion polls? *Sociological Methods & Research 54*(3), 1156–1196.

Bradley, R. A. and M. E. Terry (1952). Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika 39*(3/4), 324–345.

Bratt, C. (2025). Benefits of a pragmatic approach: Rethinking measurement invariance and composite scores in cross-cultural research. *Sociological Methods & Research*. Online first.

Broska, D., M. Howes, and A. van Loon (2025). The mixed subjects design: Treating large language models as potentially informative observations. *Sociological Methods & Research 54*(3), 1074–1109.

Buskirk, T. D., F. Keusch, L. von der Heyde, and A. Eck (2026). More parameters than populations: A systematic literature review of large language models within survey research.

Campbell, D. T. and D. W. Fiske (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin 56*(2), 81–105.

Cao, Y., H. Liu, A. Arora, I. Augenstein, P. Röttger, and D. Hershcovich (2025). Specializing large language models to simulate survey response distributions for global populations. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Albuquerque, New Mexico, pp. 3141–3154. Association for Computational Linguistics.

Chen, F., Q. Wang, L. Wang, and S. Bhatia (2026). Simulated responses from large language models are weak proxies for human economic preferences across 74 countries. *Nature Human Behaviour*.

da Silva, N. B., M. Lukk, A. Sutani, A. Moturu, H. Yang, D. Silver, M. Ratto, and T. H. Silva (2026). Population fidelity: Evaluating population representativeness in LLMs. Preprint.

Dang, T. D. A. and S. Masud (2026). Cultural value alignment via latent activation steering in large language models. ACL 2026 Student Research Workshop (non-archival track).

Davidov, E., B. Meuleman, J. Cieciuch, P. Schmidt, and J. Billiet (2014). Measurement equivalence in cross-national research. *Annual Review of Sociology 40*, 55–75.

Dettmers, T., A. Pagnoni, A. Holtzman, and L. Zettlemoyer (2023). QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, Volume 36.

Dominguez-Olmedo, R., M. Hardt, and C. Mendler-Dünner (2024). Questioning the survey responses of large language models. In *Advances in Neural Information Processing Systems*, Volume 37.

Durmus, E., K. Nguyen, T. I. Liao, N. Schiefer, A. Askell, A. Bakhtin, C. Chen, Z. Hatfield-Dodds, D. Hernandez, N. Joseph, L. Lovitt, S. McCandlish, O. Sikder, A. Tamkin, J. Thamkul, J. Kaplan, J. Clark, and D. Ganguli (2024). Towards measuring the representation of subjective global opinions in language models. In *First Conference on Language Modeling (COLM)*.

Falk, A., A. Becker, T. Dohmen, B. Enke, D. Huffman, and U. Sunde (2018). Global evidence on economic preferences. *The Quarterly Journal of Economics 133*(4), 1645–1692.

Haerpfer, C., R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, B. Puranen, et al. (2022). World values survey wave 7 (2017–2022) cross-national data-set. Madrid, Spain and Vienna, Austria: JD Systems Institute and WVSA Secretariat. Version 4.0.0.

Horton, J. J., A. Filippas, and B. S. Manning (2023, April). Large language models as simulated economic agents: What can we learn from homo silicus? Working Paper 31122, National Bureau of Economic Research. Revised February 2026.

Hu, E. J., Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Kozlowski, A. C. and J. Evans (2025). Simulating subjects: The promise and peril of artificial intelligence stand-ins for social agents and interactions. *Sociological Methods & Research 54*(3), 1017–1073.

Krsteski, S., G. Russo, S. Chang, R. West, and K. Gligorić (2026). Valid survey simulations with limited human data: The roles of prompting, fine-tuning, and rectification. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pp. 10887–10906. Association for Computational Linguistics.

Ludwig, J., S. Mullainathan, and A. Rambachan (2026). Large language models: An applied econometric framework. *Annual Review of Economics 18*, 283–316.

Lyman, A., B. Hepner, L. P. Argyle, E. C. Busby, J. R. Gubler, and D. Wingate (2025). Balancing large language model alignment and algorithmic fidelity in social science research. *Sociological Methods & Research 54*(3), 1110–1155.

Pokropek, A., T. Żółtak, E. Davidov, B. Meuleman, and P. Schmidt (2025). Challenges in multilevel modelling: Cross-group measurement noninvariance and measurement

errors. a monte carlo simulation study. *Sociological Methods & Research*. Online first.

Rafailov, R., A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn (2023). Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, Volume 36.

Santurkar, S., E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto (2023). Whose opinions do language models reflect? In *Proceedings of the 40th International Conference on Machine Learning*, Volume 202 of *Proceedings of Machine Learning Research*, pp. 29971–30004.

Steenkamp, J.-B. E. M. and H. Baumgartner (1998). Assessing measurement invariance in cross-national consumer research. *Journal of Consumer Research 25*(1), 78–107.

Sturgis, P., C. Roberts, and T. Robinson (2026). LLMs for survey pretesting: How well can large language models identify flaws in survey questions? Working Paper 15, Survey Futures.

Suh, J., E. Jahanparast, S. Moon, M. Kang, and S. Chang (2025). Language model fine-tuning on scaled survey data for predicting distributions of public opinions. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 21147–21170. Association for Computational Linguistics.

Wallach, H., M. Desai, A. F. Cooper, A. Wang, C. Atalla, S. Barocas, S. L. Blodgett, A. Chouldechova, E. Corvi, P. A. Dow, J. Garcia-Gathright, A. Olteanu, N. Pangakis, S. Reed, E. Sheng, D. Vann, J. W. Vaughan, M. Vogel, H. Washington, and A. Z. Jacobs (2025). Position: Evaluating generative AI systems is a social science measurement challenge. In *International Conference on Machine Learning, Position Paper Track*.

Wang, A., J. Morgenstern, and J. P. Dickerson (2025). Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence 7*, 400–411.

Wang, C. (2026). Silicon sampling answers with country-level assumptions, not individual attitudes: Cross-national evidence from the European Social Survey. Preprint.

World Values Survey Association (2022). World values survey time-series (1981–2022) data file. Data file version 5.0.